

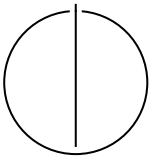
SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY —
CHAIR OF HEALTH INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis in Informatics

Decoding Music Perception: EEG-Based Genre Classification: Knowledge Base

Lukas Alex Jochim





SCHOOL OF COMPUTATION, INFORMATION AND TECHNOLOGY — CHAIR OF HEALTH INFORMATICS

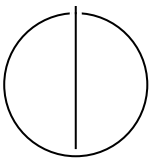
TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis in Informatics

Decoding Music Perception: EEG-Based Genre Classification: Knowledge Base

Dekodierung der Musikwahrnehmung: EEG-basierte Genre-Klassifizierung

Author:	Lukas Alex Jochim
Examiner:	Prof. Dr. Björn Schuller
Supervisor:	MSc. Simon Pistrosch
Submission Date:	01.09.2025



I confirm that this bachelor's thesis is my own work and I have documented all sources and material used.

Munich, 01.09.2025 Lukas Alex Jochim

Acknowledgments

Abstract

Contents

Acknowledgments **iv**

Abstract **v**

1 Introduction **1**
1.1 Motivation 1
1.2 Research goals 2
1.3 Methodology 2

2 Theoretical Background **3**
2.1 Medical foundations of EEG signals 3
2.2 Brain rythms 3
2.3 Correlation between music and brain activity 3
2.4 Recording and measurement of EEG signals 3
2.5 Denoising and preprocessing of EEG signals 3
2.6 Feature extraction and signal decomposition of EEG signals 3
2.7 Classification of EEG signals 3
2.7.1 Models and principles 3
2.7.2 Validation and evaluation 3

3 Related work **4**
3.1 Music Genre classification with audio features 4
3.2 EEG datasets with music stimulus 4
3.2.1 DEAP dataset 4
3.2.2 OpenMIIR initiative 4
3.2.3 NMED-T OR Music BCI 4
3.3 Emotion classification with EEG signals 4
3.4 Music Genre classification with EEG signals 4

4 Music datasets **5**
4.1 Audio Set 5
4.2 Million Song Dataset 5
4.3 MagnaTagATune 5

4.4 FMA dataset . . . 5
4.5 GTZAN Genre classification . . . 5
4.6 Final decision . . . 5

5 Methodology **6**
5.1 Acquisition . . . 6
5.1.1 General setup, environment and stimuli . . . 6
5.1.2 Acquisition device . . . 7
5.1.3 Technical setup . . . 8
5.2 Denoising . . . 8
5.3 Feature extraction . . . 8
5.4 Classification . . . 8
5.4.1 Model 1 . . . 8
5.4.2 Model 2 . . . 8
5.4.3 Model 3 . . . 8

6 Results **9**
6.1 Metric 1 . . . 9
6.2 Metric 2 . . . 9
6.3 Metric 3 . . . 9

7 Conclusion **10**
7.1 Key findings and results . . . 10
7.2 Problems and Limitations . . . 10
7.3 Future work . . . 10

Abbreviations **11**

List of Figures **12**

List of Tables **13**

1 Introduction

1.1 Motivation

The human brain is one of the most complex and important systems within the human body. The brain manages various functionalities like emotion, memory, attention and sensory processing, to name a few. Apart from that, the brain plays a significant role in how we process musical pieces. Independent of culture and age, music is a universal tool to influence one's emotions, concentration and cognition. Furthermore, the brain gives us the unique ability to retrieve many different features out of music, like melody extraction or the identification of an artist or a specific genre. Depending on the style of music, the brain reacts differently. However, although the brain is central to cognitive processes such as listening to music, the core mechanisms remain unknown and are currently black boxes.

To enlighten these black boxes, medical advancements like functional magnetic resonance imaging (fMRI) or electroencephalogram (EEG) arrived, which can measure brain activity. Especially EEG has aroused high interest since they directly measure the neural activity and can be done non-invasive with a little overhead in setup. This is why combining EEG and music has the ability to help better understand how our brain functions. By further adding means of Machine Learning (ML), a whole new interdisciplinary field opens up, bridging the gap between neurosciences and Computer science.

There is a growing research interest in the affective influence of music on the brain. However, classifying different music genres based on EEG signals remains underrepresented. This is exactly where this thesis aims to close the gap, by investigating whether it is possible to classify various musical genres depending on the EEG signals acquired using state-of-the-art ML techniques.

The research results have high potential in the domain of medical computer science. Especially with the growing investigations on Brain Machine interfaces (BMIs), adaptive music therapy, personalised music recommendations or neurofeedback systems may become possible. Additionally, entertainment technologies can become more user-centred, facilitating unique experiences that impact many sectors like education and mental health

1.2 Research goals

1.3 Methodology

2 Theoretical Background

2.1 Medical foundations of EEG signals

2.2 Brain rythms

2.3 Correlation between music and brain activity

2.4 Recording and measurement of EEG signals

2.5 Denoising and preprocessing of EEG signals

2.6 Feature extraction and signal decomposition of EEG signals

2.7 Classification of EEG signals

2.7.1 Models and principles

2.7.2 Validation and evaluation

3 Related work

3.1 Music Genre classification with audio features

3.2 EEG datasets with music stimulus

3.2.1 DEAP dataset

3.2.2 OpenMIIR initiative

3.2.3 NMED-T OR Music BCI

3.3 Emotion classification with EEG signals

3.4 Music Genre classification with EEG signals

4 Music datasets

4.1 Audio Set

4.2 Million Song Dataset

4.3 MagnaTagATune

4.4 FMA dataset

4.5 GTZAN Genre classification

4.6 Final decision

5 Methodology

5.1 Acquisition

5.1.1 General setup, environment and stimuli

The experiment has been conducted in a laboratory setting. As already mentioned in Chapter 4, the GTZAN Music dataset is chosen as a basis for the music stimuli. However, for the sake of the quality of the data, it would not be beneficial to hear all 1000 songs in one single session. To not only avoid quick attention loss during a session but also to gather as many different data under different circumstances, the whole set of audio files is partioned into 20 distinct session each lasting around 30 min. This means that per session, 50 songs are played. [1] For each of these songs, not only EEG signals are measured, but also other biosignals like PPG, IMU and temperature. However, this work will solely focus on the recorded EEG data. However, these 50 songs are not chosen randomly. To ensure that each of the ten genres from dataset is represented in each session, five songs from each genre are being played during the session. This also makes sure that there is no genre bias in the data, thus resulting in a balanced out signal across genres. During listening, the order of the songs is randomized to avoke a brain reaction that is as natural as possible as the brain cannot predict the genre that is listened to. Additonaly, to not overstimulate the brain, a five second break is implemented in between songs.

All 20 sessions have been recorded over the course of two weeks, with an average of two sessions per day. Between each session there was a short break between ten and 30 minutes. This verifies a sufficient pause of the brain to fully prepare for the next session. The test person is the author himself, a 21 year-old, right-handed college student without any common (brain) diseases. To further guarantee a high quality in the acquired signal data, the environment and setting of the experiment has to be adapted. First and foremost, the test person is instructed to sit comfortably in a chair with his eyes closed to minimize artificats coming from blinking and eye-movement. Furthermore and most importantly, any movements, especially in the head region, have to be limited and as minimal as possible since the EEG device reacts very sensitive for head movements. Apart from that, the room was quiet and isolated with dimmed

[1] Because of a corrupt audio file, one session only consists of 49 songs.

natural lighting combined with the brigthness from a Dektop monitor. The songs were played over the EEG headset (cf. Subsection 5.1.2) with average and constant volume.

5.1.2 Acquisition device

To record the EEG signal data coming from the brain, a non-traditional approach is chosen. For this study, a prototype for a hybrid biosensing platform *Hugin* by Jabra is chosen. It is a bluetooth headset that is able to play music and measure biosignals at the same time. For Audio, it uses classic bluetooth to connect to a standard PC or Laptop. For the data however, it uses a second low energy bluetooth channel for the biosignal data, which can be connected to a second data acquisition PC.

The headphones are equipped with multiple sensors and electrodes to measure biosignals like EEG, PPG, IMU and temperature. For EEG, the headphones are supplied with 2 X 4 sensor electrodes in addition to two bias electrodes. The sensors are easily configurable and provide an ultra low-noise front-end, meaning that the data points are amplified with minmal distortion and interference. Besides that, the electrodes sample with 24-bit and a frequency of 250Hz.

For PPG, there is one optical sensor on the right side, providing a high resolution of 19-bit ADC with a configurable sample rate. Apart from that, they support either raw data output or built-in algos to conclude to heart rate or SpO2.

The IMU data is sensored by a 3-axis digital accelerometer fused with a 3-axis digital gyroscope. On a full scale, the accelerometer is able to recognize acceleration from +/-2g up to +/-16g (in power of two steps). The gyroscope can measure angular velocity from $+/-125\,\mathrm{deg}/s$ up to $+/-4000\,\mathrm{deg}/s$. The recognized data then is combined together to provide accurate information on orientation, movements and position.

Last but not least, there are three digital thermometers, one on the skin, another one inside the earcup and one on the outside of the earcup to measure ambient temperature. All of those temperature sensors have an accuracy of +/- 0.1C.

Since this headset still is a prototype, they still is ongoing development and research on the headset. Nonetheless, there already were some small-scale research projects employing the biosignals from the headphones. These projects range from stress detection using the PPG data to device control and the recognition of cognitive fatigue using the EEG sensors.

5.1.3 Technical setup

5.2 Denoising

5.3 Feature extraction

5.4 Classification

5.4.1 Model 1

5.4.2 Model 2

5.4.3 Model 3

6 Results

6.1 Metric 1

6.2 Metric 2

6.3 Metric 3

7 Conclusion

7.1 Key findings and results

7.2 Problems and Limitations

7.3 Future work

Abbreviations

List of Figures

List of Tables